Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Demo-oriented engineer with **2+ years of experience** building relational workflow systems, customer-facing interfaces, automations, and proof-of-concept platforms that translate complex business processes into intuitive, reusable product experiences.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed multi-tenant relational schemas across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+ records**, creating reusable data models that reduced dashboard latency by **71%**.
- Built AI-enabled workflow automation and dashboard experiences with **FastAPI**, cloud storage, and background orchestration, supporting **5K+ monthly jobs** with **99%+** reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Built a polished **React** and **Flask** interface for question upload, grading history, and review workflows, supporting **150+ instructors** and **10K+ requests/day**.
- Designed reusable automation flows for rubric generation, secure file ingestion, and model-driven evaluation, reducing turnaround time by **60%** while keeping the experience adaptable to multiple use cases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built cross-platform microservice workflows with reusable modules, automated validation, and structured documentation, improving response times by **30%** and pipeline efficiency by **68%**.
- Maintained CI-backed delivery and regression processes using **Jenkins, PyTest, and Mockito**, helping turn complex backend processes into repeatable internal tooling for distributed teams.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN**-based business operations platform with **8** services for authentication, resource allocation, billing, permissions, notifications, and analytics, synchronizing **2M+ records**.
- Modeled structured data and operational logic in **Azure SQL**, then paired it with user-facing workflows and analytics, reducing resource wastage by **42%** and increasing lead conversions by **15%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed the end-to-end workflow for a two-sided booking platform, mapped **12+ backend integration points**, and presented the solution to the client before implementation to align on the product story and operational logic.
- Built **React/TypeScript** interfaces and **Flask** APIs for pricing, booking, coupon, and payment workflows over **50K+ records**, with webhooks and analytics supporting sub-**200ms** user flows.

Skills

Workflow & Data Design: Relational Schemas, Business Logic Modeling, Linked Workflow Design, Permissions, Automation Logic, Documentation
Frontend & Interfaces: React, Next.js, React Native, TypeScript, JavaScript, TailwindCSS, Dashboard Design
Backend & Integrations: Flask, FastAPI, Node.js, Express.js, REST APIs, Webhooks, SQL, PostgreSQL, MySQL, Azure SQL, Firebase
Engineering: AI-Assisted Workflows, Proof-of-Concept Builds, CI/CD, Docker, AWS, Customer-Facing Technical Storytelling

Projects

HMIS – Hospital Management Information System Live | *TypeScript, Appwrite, Twilio, Sentry*
- Built a proof-of-concept healthcare workflow environment with appointment logic, consent-based record access, and stakeholder-specific dashboards, supporting **100+ daily appointments** while reducing administrative workload by **30%**.

Big-Brain Live | *PostgreSQL, REST APIs, Collaborative Workflows*
- Built an AI-powered collaborative workspace for **200+ users** with structured document workflows, secure access controls, and retrieval over **50K+ records**, reducing review time by **60%**.

Chat2SQL GitHub | *TypeScript, PostgreSQL, Docker*
- Built a structured data workspace supporting **10+** databases with schema exploration, SQL generation, and reusable multi-service workflows, mirroring the kind of relational proof-of-concept environments used in technical demos.